

Automated Deep Learning-Based Segmentation and Quantitative Thickness Analysis of OCT Skin Layers in Diabetic Foot Patients

Abstract

The optical coherence tomography (OCT) imaging technique can give high-resolution noninvasive cross-sectional images of skin microstructures without performing any biopsy of the tissue. hence, it is feasible for use in clinical practice as a diagnostic method in dermatology. For diabetes mellitus individuals and more specifically, those who develop diabetic foot ulcers (DFU), there is an established relationship between changes in the microstructure of their skin and the likelihood of developing an ulcer or poor wound healing. In this research, a novel deep learning semantic segmentation algorithm is developed to automatically segment the three key plantar skin structures, namely, the stratum corneum (SC), epidermis (ED), and dermis (DR).

The dataset comprised 723 OCT images with corresponding pixel-wise segmentation masks: 397 from healthy controls and 326 from DFU patients. Which was partitioned into training (Healthy: 348, DFU: 294), validation (Healthy: 33, DFU: 19), and test (Healthy: 16, DFU: 13) subsets. A total of 11 deep learning architectures were considered in this paper, which of 9 is built upon a unified custom Boundary-Aware Residual Squeeze U-Net (BARS-UNet) baseline, with individual architectural modifications inspired by U-Net (ResNet34), U-Net++ (ResNet34, EfficientNet-B4), DeepLabV3+ (ResNet34), Attention U-Net (ResNet50), Residual U-Net (ResNet34), Dense U-Net (DenseNet121), U-Net (EfficientNet-B4, VGG-16) along with two predefined architectures Swin-UNet (Swin-Tiny) [9] and MedSAM-UNet (SAM ViT-B) [10] included for external benchmark comparison. Every single one of the 9 custom models has residual connections, squeeze-and-excitation channel attention, and even a dedicated boundary-aware feature branch, meaning that no pretrained encoder backbones are required.

The pre-processing steps for the signal processing algorithm consisted of lateral Gaussian smoothing, CLAHE, and unsharp masking and were performed uniformly on all experiments. Thickness of the individual layers was estimated based on the morphological determination of upper and lower boundaries of the layers in each column in the image followed by lateral Gaussian smoothing of the obtained profiles, rather than simple counts of pixels in each column. Comparisons between cohorts relied on the Welch's t-test, Shapiro-Wilk normality test, Cohen's d, and bootstrapping. Gradient-weighted Class Activation Mapping (Grad-CAM) was applied to the best-performing model to provide visual explainability of model attention.

The best performing model for DFU segmentation was the Residual U-Net (ResNet34), which is an advanced implementation of BARS-UNet with residual scaling, demonstrating a test-

set Dice score of 0.8518 and IoU of 0.7554. The thickness analysis demonstrated that there was a statistically significant increase in epidermis thickness (Healthy: 15.871 ± 5.837 px; DFU: 30.798 ± 10.773 px; $p = 0.0509$; Cohen's $d = 1.723$, large effect) and a statistically significant decrease in dermis thickness (Healthy: 37.343 ± 4.81 px; DFU: 26.750 ± 6.060 px; $p = 0.0055$; Cohen's $d = -1.9363$, large effect). There were no statistically significant differences in stratum corneum thickness ($p = 0.3174$). Logistic regression classifier employing thickness estimates resulted in an AUC-ROC of 0.9567, while five-fold cross-validation verified the stability of the models with an average DFU Dice of 0.7909 ± 0.0130 .

The proposed framework provides a scalable, reproducible, and clinically interpretable tool for quantifying microstructural skin changes in diabetic foot patients, with possible application to early detection, risk stratification, and clinical screening workflows.

Keywords: *Optical Coherence Tomography, Diabetic Foot Ulcer, Deep Learning, Skin Layer Segmentation, BARS-UNet, Boundary-Based Thickness Analysis, Custom Architecture, Swin-UNet, MedSAM-UNet, Grad-CAM*

1. Introduction

Diabetes mellitus is one of the commonest chronic metabolic disorders globally, with about 537 million adults suffering from diabetes mellitus as at 2021, with future projections suggesting that this number will increase to more than 783 million by 2045 [1]. There are many types of complications in diabetes mellitus, with considerable microvascular and neurodegenerative changes being implicated in their pathophysiology. nonetheless, one of the most dangerous forms of complications in diabetes mellitus is DFU. Around 15% to 25% of people living with diabetes mellitus are reported to have DFU in their lifetime, and it is also one of the leading reasons for nontraumatic limb amputation [2]. Prebiome of ulcers is characterized by structural modifications, including epidermal thickening and dermal thinning and degeneration in layers separation, which usually happen prior to ulcer formation [3].

The optical coherence tomography (OCT) system represents a noninvasive interferometric technique for acquiring high-resolution cross-sectional images of biological tissue structures at micrometer scale resolution without having to conduct any tissue biopsy. For skin applications, OCT allows direct observation of the stratum corneum, epidermis, and dermis, with the axial resolutions of about 3–5 μm [4], which provides adequate resolution for morphological analysis. Unlike histopathologic methods, OCT provides an ability to carry out repeatable measurements of patients without posing a risk to them; thus, this approach is particularly suitable for studying diabetes mellitus associated with the feet. Previous research has already proven applicability of OCT for assessing skin thickness in diabetic patients, albeit mostly using manual or semiautomatic analysis, which is quite laborious and prone to observer bias [5].

Convolutional neural network (CNN)-based methods have played an important role in medical image segmentation through pushing forward the frontier of knowledge. The introduction of U-Net [6], with its encoder-decoder structure and skip connections, established a foundational framework for biomedical segmentation. Subsequent architectures such as U-Net++ [7], DeepLabV3+ [8], and transformer-based models including Swin-UNet [9] have further extended segmentation capabilities for complex medical imaging tasks. Foundation models such as MedSAM [10] have more recently demonstrated strong generalization across diverse medical imaging domains. Despite this progress, automated multi-layer segmentation of OCT images specific to diabetic foot pathology remains an underexplored area, and quantitative thickness comparisons between healthy and DFU cohorts using deep learning predictions have rarely been reported.

Despite this progress, several important gaps remain in the automated analysis of OCT skin images specifically for diabetic foot pathology. First, studies that use pre-trained encoder backbone networks depend on the feature distribution in ImageNet, which is not necessarily ideal due to coherent speckle noise and contrast dependency present in OCT images [11,12]. Second, traditional methods that use the simple counting of pixels in columns to determine layer thickness do not take into account the effect of boundary noise, segmentation mistakes, and irregularity of borders, which is prevalent in DFU skin specimens [13]. Third, systematic comparisons of diverse architectural design choices particularly within a controlled, unified framework that eliminates backbone variability have rarely been reported for this specific task.

These gaps are explored by means of an architecture-specific design of custom BARS-UNets built uniformly from a shared BARS-UNet baseline in which all nine networks are characterized by identical architectural traits - residual blocks, squeeze-and-excitation channel-wise attention, and a boundary-aware feature path, but differ in one key concept. While two predefined architectures, Swin-UNet [9,21] and MedSAM-UNet [10], are included as external benchmark comparators leveraging large-scale pretrained representations. This design allows direct evaluation of custom architecture design choices alongside the generalization capability of foundation-scale pretrained models.

The main working include: (i) Comparative studies on the performance of nine customized BARS-UNet and two predefined models segmentation approaches individually trained on OCT scans from both healthy individuals and people suffering from DFUs without utilizing any pre-trained backbone model for custom models; (ii) Signal processing approach to pre-process the OCT scan images in order to reduce noise and improve boundaries of the images; (iii) Boundary-based thickness measurement procedure that involves morphological boundary extraction followed by the use of Gaussian curve fitting; (iv) Statistical tests like Welch's test, Cohen's d, Shapiro-Wilk test for normality, and bootstrapping; (v) Grad-CAM explainability analysis of the

best-performing model to confirm clinically relevant attention patterns; (vi) Five-fold cross validation and logistic regression for classification based on AUC-ROC.

2. Methodology

2.1 Dataset and Annotation

A pair of separate OCT image datasets was used: the first dataset comprised plantar skin images of healthy subjects, and the second dataset comprised images of plantar skin of DFU patients. The datasets were divided into training, validation, and testing datasets as follows: Healthy: 348 training, 33 validation, and 16 test; DFU: 294 training, 19 validation, and 13 test. All OCT images were normalized to have dimensions of 512×512 pixels. Images were acquired from the plantar surface of the foot, targeting the anatomical regions most susceptible to ulceration in diabetic patients.

Annotations based on polygonal manual delineation were done for three categories of tissue: stratum corneum (SC; label = 1), epidermis (ED; label = 2), and dermis (DR; label = 3), while label 0 corresponds to background. Annotation was done using a structured web-based labeling interface with two projects for each group: a healthy cohort and DFUs. Annotation was conducted such that there was uninterrupted vertical boundary coverage along the entire width of the image, which made it possible to analyze boundary columns one at a time, while densely sampling polygon points in order to reflect the complexity of boundaries of DFU-affected tissue. Then these annotations were exported in COCO segmentation format [26] and converted into pixel-level multi-class label masks saved in lossless PNG format to prevent compression artifacts.

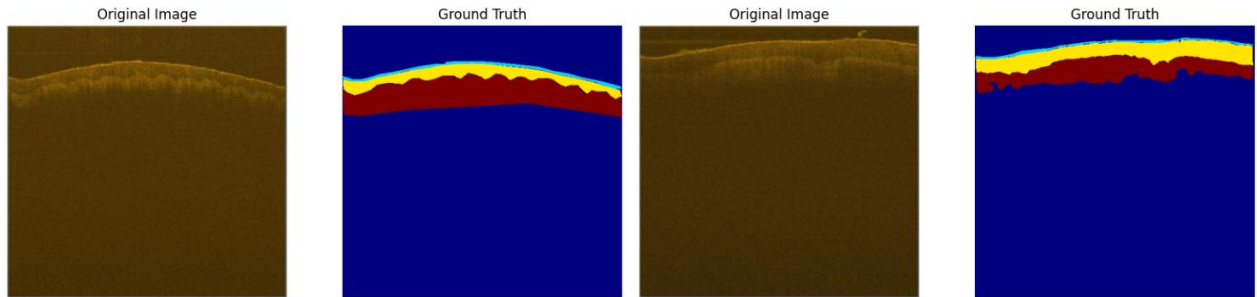


Figure 1. Representative OCT images from the healthy control (left) and DFU (right) cohorts alongside their corresponding ground truth segmentation masks. Note visible boundary disruption and irregular layer morphology in the DFU sample.

2.2 Signal Processing Preprocessing Pipeline

The same five-step procedure of signal processing was applied to each input image prior to training and inference. The intention behind this pipeline was to suppress coherent speckles in the OCT images and enhance contrast between the layers. The stages are as follows:

- **Grayscale Conversion:** Input RGB images were converted to single-channel grayscale to reduce channel redundancy and concentrate subsequent processing on intensity-based structural information relevant to OCT depth profiles.
- **Lateral Gaussian Smoothing ($\sigma = 1$):** One-dimensional Gaussian filter (with $\sigma = 1$) was used to filter along the lateral (B-scan) axis by means of `scipy.ndimage.gaussian_filter1d`. The effect of such filtering is to suppress high frequency speckle noise while keeping the structural envelope in the axial direction of each A-scan unchanged, which is necessary for precise boundaries localization.
- **Contrast Limited Adaptive Histogram Equalization (CLAHE):** The process of Contrast limited adaptive histogram equalization was used where clip limit equals to 2.0 while grid tile size was 8×8 pixels. It is important to note that this method allows to enhance local contrast especially at the border between epidermis and dermis but without increasing noise level in spatially homogenous regions [23]. It is necessary to mention that Contrast limited adaptive histogram equalization is an ideal way for OCT image processing since there are no speckle statistics constancy in the OCT images and global histogram equalization will increase signal level too much.
- **Unsharp Masking ($\sigma = 3$):** Unsharp masking was done through subtracting a Gaussian-blurred copy of CLAHE-improved OCT image from their weighted combination where weight of the blurred image is -0.5 and weight of the original OCT image is 1.5. This step sharpens layer boundaries and improves edge definition, which is particularly beneficial for the frequently disrupted epidermal boundaries found in DFU cases.
- **Resize and Channel Conversion:** Grayscale image was reconverted into three-channel format BGR, scaled to 512×512 pixels through bilinear interpolation, after that converted to RGB image with normalization.

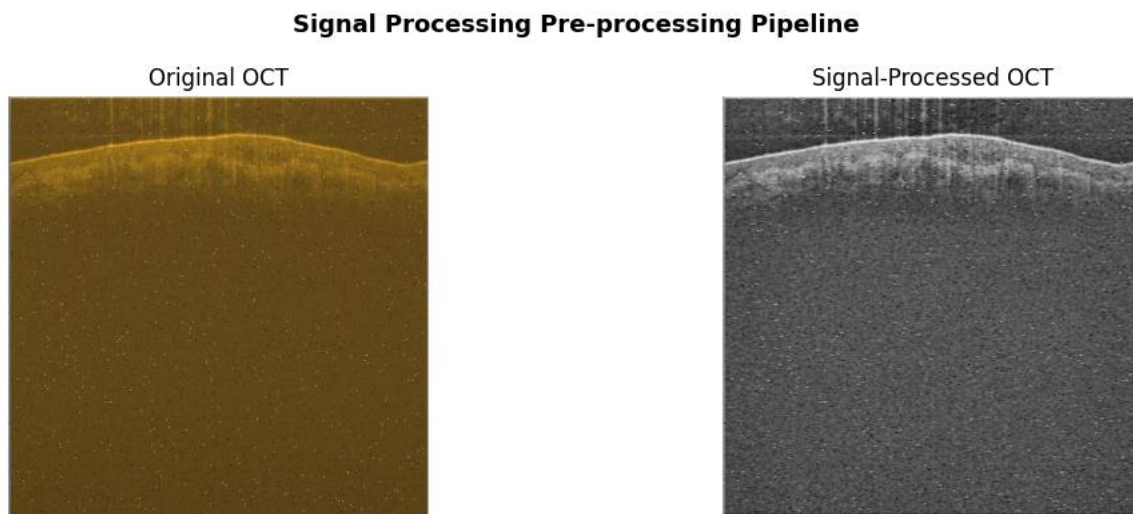


Figure 2. Original OCT image (left) and signal-processed OCT image (right) after the full preprocessing pipeline. The processed image demonstrates improved inter-layer boundary contrast and reduced speckle noise.

2.3 Boundary-Based Layer Thickness Measurement

Using a boundary-based approach the thickness of the layers was measured based on the predictions from the segmented masks. The main difference between the boundary-based approach and other layer-thickness measurement approaches, such as those that simply count pixels in each image column by class label value, is that the former is based on detecting the boundaries of the layers.

The thickness measurement process follows three main steps. In the first step, for each tissue type (SC, ED, DR), a binary layer mask is produced, which is subsequently morphologically eroded with a 1×3 vertical structuring element, resulting in identification of internal pixels. Then, the difference between the binary mask and the eroded mask is computed as boundary, resulting in only the outer pixels comprising the boundary. Second, for each image column, the topmost and bottommost non-zero pixel rows in the original binary layer mask are recorded, and thickness for that column is defined as the vertical distance between these boundary positions plus one pixel. Columns of zero thickness (absence of the layer) are not considered. Lastly, one-dimensional Gaussian smoothing ($\sigma = 2$) in the lateral direction is performed independently on the top and bottom boundary maps, and thicknesses (difference between bottom and top boundary) averaged across all valid columns are computed. As an additional technique for verification, gradient-based boundary detection using vertical Sobel filter and sub-pixel boundary determination based on Gaussian center-of-mass estimation were also implemented.

2.4. Custom BARS-UNet Architecture Framework

Nine segmentation network architectures considered in this work, which was implemented based on the BARS-UNet architecture, which was developed from scratch and does not use any pretrained encoders. The BARS-UNet baseline incorporates four structural properties uniformly applied across all variants:

- **Residual Connections:** Residual connections are utilized in each convolutional layer; it can be described by the formula $\text{Output} = F(x) + \text{shortcut}(x)$. In case when there is a difference between the number of input and output channels, there is 1×1 convolution and batch normalization within shortcut. This supports stable gradient flow and reduces vanishing gradient risk in deeper networks [6].
- **Squeeze-and-Excitation (SE) Channel Attention:** SE channel recalibration technique is used in all convolutional layers [29]. First, global average pooling is performed in order to reduce the dimensionality of feature map into channel descriptors. After that, fully-connected network with 16 layers and with ReLU and Sigmoid activations is used. Coefficients for channel recalibration are calculated based on this layer. The resulting channel weights are multiplied element-wise onto the feature map, enabling the network to amplify informative channels and suppress uninformative ones.
- **Boundary-Aware Feature Branch:** Apart from the semantic decoder branches, there is a parallel boundary aware feature branch, where boundary-aware features are obtained through depthwise 3×3 convolution, channel projection 1×1 , batch normalization, and

ReLU activation. After obtaining the features from the boundary branch, these are combined with the semantic decoding feature map using 1×1 convolution, thereby making it possible to leverage the boundaries while performing segmentation, particularly for OCT segmentation of thin layers.

- **Enhanced Feature Fusion:** In the decoder, after fusing the bottleneck feature with the encoder feature via residual block, the feature map is fused with the boundary branch output via a second 1×1 convolution. This two-stage fusion preserves both global context and local boundary detail.

The encoder follows a four-stage downsampling design (channel progression: $3 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 512$) using max pooling between stages, and the bottleneck expands to 1024 channels. The decoder is similar to this structure with transposed convolutions for upsampling. All nine variants share this backbone and differ only in one targeted architectural modification, described below.

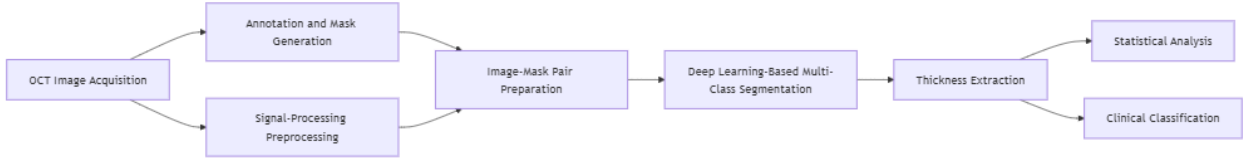


Figure 3. Workflow of the proposed automated OCT skin layer segmentation and quantitative thickness analysis framework. OCT images undergo annotation and preprocessing before preparation of the image-mask dataset for deep learning-based segmentation, followed by thickness extraction, statistical analysis, and clinical classification

2.4.1 Architectural Variants

The nine custom variants are described below. Each one incorporates a single change to the base architecture, the BARS-UNet, to ensure a fair architectural comparison:

- **Model 1 - U-Net (ResNet34):** Implements the BARS-UNet baseline without additional modification. considered as the reference model for all comparative analyses.
- **Model 2 - U-Net++ (ResNet34):** Adds nested skip refinement modules between encoder skip connections and the decoder. Refinement modules consist of two more convolution layers combined with the SE attention mechanism and a shortcut connection that connects directly to the skip connections, similar to the idea of dense skip connection aggregation used in UNet++ [7].
- **Model 3 - U-Net++ (EfficientNet-B4):** Replaces the standard single-layer SE block with a deeper two-layer FC squeeze-and-excitation module (DeepSEBlock) throughout all encoder, bottleneck, and decoder components. The DeepSEBlock adds an intermediate fully connected layer between the squeeze and excitation stages, providing stronger channel recalibration capacity inspired by the compound scaling philosophy of EfficientNet [20].
- **Model 4 - DeepLabV3+ (ResNet34):** Replaces the standard bottleneck with a dilated pyramid bottleneck implementing parallel atrous convolutions with dilation rates of 1, 2, 4,

and 8, summed elementwise and projected back to 1024 channels. Thus, each one allows for multi-scale receptive field learning in the bottleneck layer where features can capture information at different scales without increasing spatial dimensions [8].

- **Model 5 - Attention U-Net (ResNet50):** Augments skip connections with additive spatial attention gates that modulate the encoder feature maps prior to concatenation with the decoder. Each gating layer computes a spatial attention weight by first computing the additively learned compatibility score of the gating input from the decoder feature and the corresponding input from the encoder feature using a sigmoid function [22].
- **Model 6 - Residual U-Net (ResNet34):** Applies scaled residual learning throughout: the residual shortcut contribution is multiplied by a fixed scaling factor $\alpha = 0.5$, yielding output $= F(x) + 0.5 \cdot \text{shortcut}(x)$. As residual connections have been implemented in the BARS-UNet baseline, this version pushes residual connectivity further through deep residual stacks and different skip concatenations in order to maintain distinct architectural differences. The residual connections are enhanced through scaling and employed throughout the encoder and decoder layers.
- **Model 7 - Dense U-Net (DenseNet121):** Applies dense feature concatenation within each convolutional block: $x_1 = \text{Conv}_1(x)$, $x_2 = \text{Conv}_2(x_1)$, combined = $\text{Proj}(\text{cat}(x_1, x_2)) + \text{shortcut}(x)$. Dense connection inside the block promotes explicit feature re-utilization at every computational step, following the DenseNet concept of connecting every layer to all other layers that follow within a dense block [19].
- **Model 8 - U-Net (EfficientNet-B4):** Replaces standard 3×3 convolutions with depthwise separable convolutions, a spatial depthwise convolution followed by a pointwise 1×1 projection to substantially reduce the parameter count and computational cost. Both residual and SE blocks remain the same, although the depthwise structure decreases the interactions between channels within a layer, leading to an extremely light architecture like those developed for mobile devices [20].
- **Model 9 - U-Net (VGG-16):** Increases network depth by stacking four sequential convolutional layers within each encoder block and two in each decoder block, without residual shortcuts inside individual blocks (VGG-style plain convolutions [30]). SE attention and boundary branch are kept from the BARS-UNet baseline while the lack of intra-block shortcuts allows depth through consecutive feature transformations rather than through residual mappings.

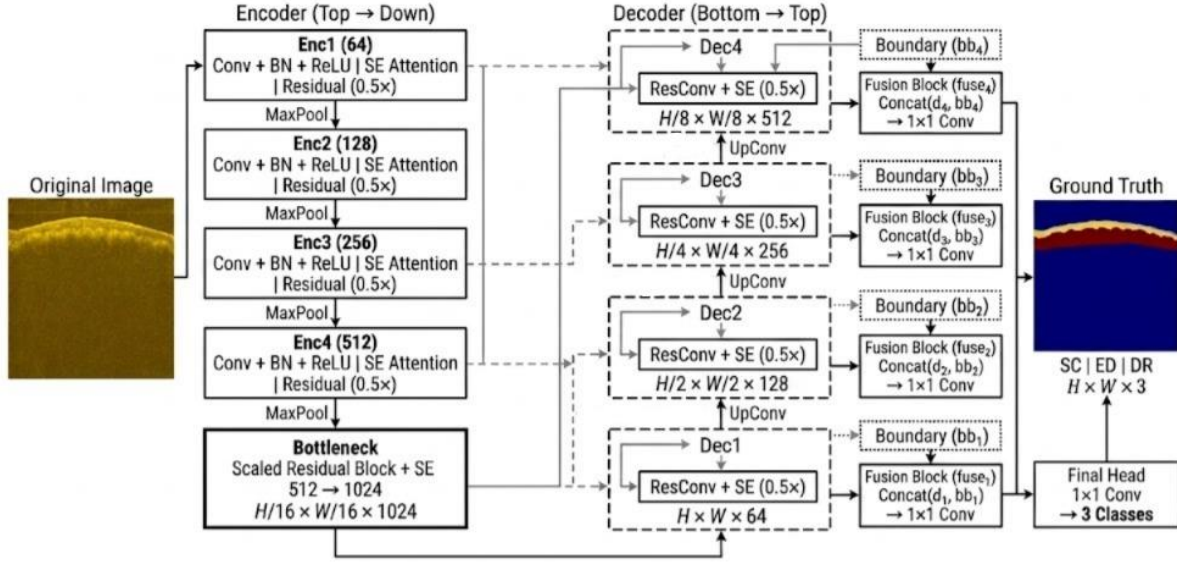


Figure 4. BARS-UNet baseline architecture. ResConv = residual convolutional block, SE = squeeze-and-excitation channel attention, BB = boundary-aware feature branch, fuse = feature fusion. Shared uniformly across all nine variants.

2.4.2 Predefined Architectures

In addition to the nine custom BARS-UNet variants, two predefined architectures were included to benchmark performance against established models leveraging large-scale pretrained visual representations:

- Model 10 - Swin-UNet (Swin-Tiny):** A transformer-based segmentation architecture utilizing the Swin Transformer encoder [9,21] with shifted window self-attention mechanisms that capture long-range spatial dependencies across the image. Input images were internally resized to 224×224 for the Swin-Tiny encoder (pretrained on ImageNet-1K), with the decoder upsampling feature representations back to 512×512 via a lightweight four-stage UNet decoder with skip connections (channel dimensions: [96, 192, 384, 768]). This architecture is included to evaluate whether transformer-based long-range context modeling offers advantages over local convolutional approaches for OCT layer segmentation.
- Model 11 - MedSAM-UNet (SAM ViT-B):** Utilizes the Segment Anything Model (SAM) Vision Transformer-B (ViT-B) image encoder [10], pretrained on the SA-1B large-scale dataset and initialized with MedSAM weights, combined with a lightweight four-stage UNet decoder for multi-class segmentation. The SAM encoder produces a 256-channel embedding at 64×64 spatial resolution from 1024×1024 input, which is progressively upsampled ($64 \rightarrow 28 \rightarrow 56 \rightarrow 128 \rightarrow 256 \rightarrow 512$) through the decoder. The encoder is frozen during training, and only the decoder layers are fine-tuned. By leveraging large-scale pretrained visual representations, MedSAM-UNet captures rich contextual information and supports improved delineation of thin and low-contrast tissue

layers in OCT images. The high inference time (3492 ms) reflects the computational cost of SAM's ViT-B encoder and necessitates server-level GPU deployment.

These two predefined models are trained with the same preprocessing pipeline, loss function, optimizer, and training protocol as the custom models to ensure fair evaluation. Their inclusion enables direct comparison between domain-specific custom architectures and foundation-scale pretrained models on this specialized OCT segmentation task.

2.5 Training Protocol and Memory Management

All models were trained with a combination of Dice loss and cross-entropy loss of equal weight. The loss is optimized with respect to region overlap and pixel-wise classifications. The Adam optimizer [27] was used with an initial learning rate of 1×10^{-4} and a cosine annealing learning rate schedule over 50 epochs. The standard training batch size was 4, applied to the majority of models on an NVIDIA Tesla T4 GPU (15.6 GB VRAM).

Three models, which are architecturally more advanced U-Net++, Attention U-Net and Dense U-Net, demanded an alternate memory allocation strategy as their memory requirements were higher per one forward pass operation. The above models were trained with a physical batch size of 1 and with the use of gradient accumulation over 4 passes, thus providing an effective batch size of 4. In this manner, it can be ensured that both the gradients and dynamics would be statistically equivalent to those obtained from the model which was trained with the use of a physical batch size of 4, maintaining fairness in comparative evaluation. Swin-UNet and MedSAM-UNet were trained with the same protocol (BS=4, 50 epochs, same loss and optimizer) to ensure a fair comparison. The effective batch size of 4 is preserved consistently across all models.

2.6 Evaluation Metrics

Performance on the test datasets was quantified using conventional per-pixel metrics independently calculated for each class and averaged over all non-background classes. Metrics were calculated for each individual class based on results from all test cases and presented as means. Letting TP, FP, and FN denote true positives, false positives, and false negatives for a given class:

Dice Coefficient = $\frac{2TP}{2TP+FP+FN}$, measures region-level overlap and is the primary performance indicator.

Intersection over Union (IoU) = $\frac{TP}{TP+FP+FN}$, captures the spatial agreement between prediction and ground truth.

Precision = $\frac{TP}{TP+FP}$, quantifies the fraction of predicted pixels that are correctly classified.

Recall = $\frac{TP}{TP+FN}$, quantifies the fraction of true class pixels successfully detected.

Per-class metrics were computed for SC, ED, and DR independently. Mean values across classes are reported as the aggregate model performance. Inference time was measured using synchronized GPU timing to characterize deployment suitability.

2.7 Explainability Analysis

For the visual interpretation of the model's decision process, Gradient-weighted Class Activation Mapping (Grad-CAM) [31] was utilized on the highest-performing model of Residual U-Net (ResNet34). Grad-CAM is an approach which calculates the gradient of the output score with regard to the activations of a chosen convolutional layer and conducts the global average pooling of the gradients to construct class-specific localization maps. Those saliency maps are overlaid on the input OCT images, which allows highlighting the areas of interest that the model attends to in order to make predictions for the skin layer classes.

In this work, Grad-CAM mappings were constructed separately for each of the three target classes (SC, ED, DR) for representative images of both groups of healthy controls and DFUs. A composite explainability score was calculated based on the spatial correlation of the model predictions with the corresponding Grad-CAM saliency maps. Such a score quantitatively evaluates the degree of model attention being concentrated on anatomically relevant areas. The Residual U-Net (ResNet34) model reached a composite Grad-CAM score of 0.8518 for the DFU images, indicating that the model's attention was correctly directed at the boundaries of skin layers instead of image backgrounds.

2.8 Statistical Analysis

Statistical comparison of layer thickness distributions between healthy and DFU cohorts followed a systematic approach. First, the Shapiro-Wilk test [24] was performed to check for the normality of distribution of thicknesses within each category per layer. Welch's two-sample t-test (for unequal variance) was applied to test group means with significance level set at $\alpha = 0.05$. Effect size was computed as Cohen's d [25], defined as the mean difference divided by the pooled standard deviation, and interpreted as: $|d| < 0.2$ = negligible, $0.2-0.5$ = small, $0.5-0.8$ = medium, $|d| \geq 0.8$ = large. Bootstrap 95% confidence intervals for the mean difference (DFU minus healthy) were estimated based on 2000 bootstrap samples using stratified sampling.

Cross-validation using fivefold stratified sampling was applied to the DFU training and validation datasets. Logistic regression classification model based on the thickness feature vector

(SC, ED, DR) was assessed clinically for its ability to discriminate between healthy subjects and patients with DFUs using the AUC-ROC.

3. Results and Discussion

3.1 Comparative Segmentation Performance

Table 1 shows the results of segmentation of all 11 architectures on healthy and DFU cohorts. Among the nine custom variants, both groups demonstrate competitive dice scores due to effectiveness of the BARS-UNet framework and preprocessing steps used. In the healthy group, the best dice score belongs to Residual U-Net (ResNet34) with the result of 0.8998, second best is U-Net (VGG-16) with dice score 0.8992 and third is U-Net (ResNet34) with 0.8988. In the DFU group, once again the best result in terms of Dice score was observed in the case of Residual U-Net (ResNet34) with the dice score of 0.8518 and IoU of 0.7554, followed by DeepLabV3+ (ResNet34) with 0.8464 and U-Net (VGG 16) with 0.8452.

The consistently lower DFU performance across all models relative to the healthy cohort reflects the increased structural complexity and boundary irregularity characteristic of diabetic plantar skin. Using the scaled residuals in the Residual U-Net approach creates greater gradients and assists in better feature distinction, this appears to be an effective technique for distinguishing structures containing disruptions as seen in the DFU images. The Dense U-Net scored the worst on DFU Dice among all the variants at 0.815. Although the architecture incorporates dense connectivity within convolutional blocks to promote feature reuse, this design can lead to excessive feature aggregation and reduced feature selectivity. In the context of DFU segmentation, where accurate boundary localization is critical, such dense concatenation may introduce redundant or less discriminative features, thereby weakening the model’s ability to precisely delineate irregular lesion edges. All performance differences should be interpreted in the context of the limited test set size [28].

Table 1. Segmentation performance of all nine custom BARS-UNet variants and Swin-UNet and MedSAM-UNet on healthy and DFU test sets.

Model	H-Dice	H-IoU	H-Prec	H-Rec	DFU-Dice	DFU-IoU	DFU-Prec	DFU-Rec
U-Net (ResNet34)	0.8988	0.8183	0.8937	0.9118	0.844	0.7404	0.8275	0.8827
U-Net++ (ResNet34)	0.8864	0.7991	0.899	0.8849	0.8366	0.7378	0.8277	0.8634
U-Net++ (EfficientNet-B4)	0.8943	0.811	0.8974	0.8995	0.8343	0.7346	0.8432	0.8434
DeepLabV3+ (ResNet34)	0.8964	0.8142	0.8901	0.9109	0.8464	0.7487	0.846	0.8717

Attention U-Net (ResNet50)	0.8959	0.8138	0.8949	0.9055	0.8299	0.7305	0.8168	0.864
Residual U-Net (ResNet34) *	0.8998	0.8201	0.9013	0.9054	0.8518	0.7554	0.8477	0.8771
Dense U-Net (DenseNet121)	0.8806	0.7893	0.8987	0.874	0.815	0.7091	0.8103	0.8601
U-Net (EfficientNet-B4)	0.8976	0.8162	0.8907	0.9117	0.8428	0.7428	0.8435	0.8661
U-Net (VGG-16)	0.8992	0.8187	0.8879	0.9175	0.8452	0.7448	0.8292	0.8847
Swin-UNet (Swin-Tiny)	0.8816	0.7908	0.8569	0.9171	0.8166	0.7084	0.7944	0.8729
MedSAM-UNet (SAM ViT-B)	0.8959	0.8132	0.8820	0.9177	0.8396	0.7345	0.8127	0.8928

*H = Healthy cohort; DFU = Diabetic foot ulcer cohort. * = Best-performing model on DFU Dice. Models are custom BARS-UNet variants, not pretrained architecture implementations.*



Figure 5. Dice score comparison across all eleven evaluated models on healthy (left) and DFU (right) test sets.

3.2 Per-Class Segmentation Performance

Per-class segmentation metrics for the model with the best overall accuracy, Residual U-Net (ResNet34), are reported in Table 2 for both cohorts. In the healthy cohort, the epidermis received the top Dice result of 0.9204, indicative of the highly defined and continuous nature of this layer in the healthy plantar foot. The dermis received 0.8958 Dice and the stratum corneum 0.833. For the DFU group, the epidermis also received a relatively high Dice of 0.8916 while the dermis received the lowest value among all classes and cohorts of 0.7930.

This reduction in dermal segmentation performance carries direct clinical significance. The dermis in DFU patients exhibits structural disorganization including reduced collagen density, irregular boundary morphology, and fragmented tissue architecture, all of which challenge automated boundary localization. The DFU stratum corneum Dice result of 0.8707 is indicative of the relatively well-preserved nature of the outermost layer in terms of segmentation, which has been observed to maintain its boundaries up to later stages of disease development [16].

Table 2. Per-class segmentation performance of Residual U-Net (ResNet34) on healthy and DFU test sets.

Cohort	Layer	Dice	IoU	Precision	Recall	Accuracy
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Healthy	SC	0.8833	0.7914	0.8912	0.8793	0.9978
Healthy	ED	0.9204	0.8538	0.9125	0.9305	0.9950
Healthy	DR	0.8958	0.8150	0.9002	0.9064	0.9845
DFU	SC	0.8707	0.7719	0.8546	0.8932	0.9976
DFU	ED	0.8916	0.8213	0.9378	0.8715	0.9884
DFU	DR	0.7930	0.6730	0.7508	0.8666	0.9810

SC = Stratum Corneum, ED = Epidermis, DR = Dermis..

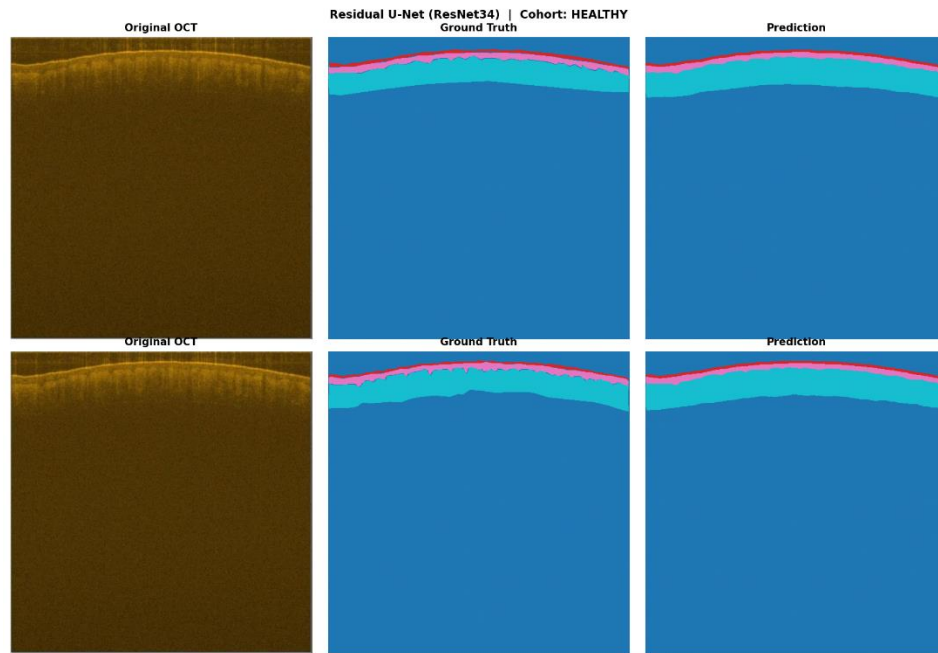


Figure 6. Segmentation results of Residual U-Net (ResNet34)inspired custom UNet: original OCT image (left), ground truth mask (center), and predicted segmentation mask (right) for representative images from the healthy cohorts. Color coding: red = stratum corneum, pink = epidermis, light blue = dermis, blue = background.

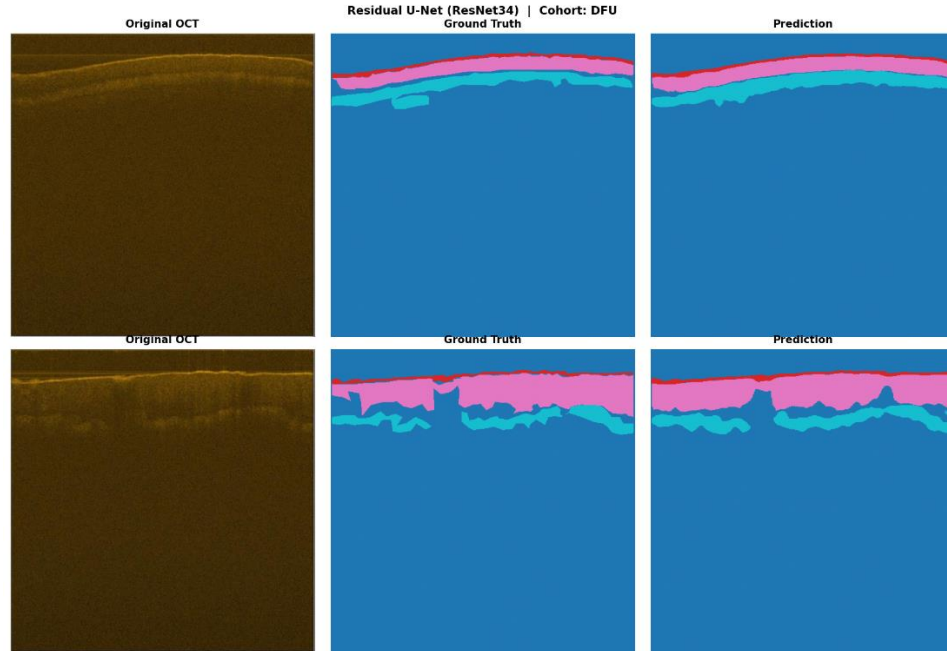


Figure 7. Segmentation results of Residual U-Net (ResNet34): original OCT image (left), ground truth mask (center), and predicted segmentation mask (right) for representative images from the DFU cohorts. Color coding: red = stratum corneum, pink = epidermis, light blue = dermis, blue = background.

3.3 Quantitative Boundary-Based Thickness Analysis and Statistical Comparison

Layer thickness estimation using boundary detection was performed on images segmented by the best-performing Residual U-Net inspired custom U-Net model and then statistically analyzed. Table 3 summarizes the mean thickness, standard deviation, Shapiro-Wilk normality test results, Welch's t-test statistic and p-value, Cohen's d effect size, and 95% bootstrap confidence interval for each skin layer.

Table 3. Statistical comparison of skin layer thickness between healthy and DFU cohorts using boundary-based measurement (Residual U-Net inspired custom U-Net predictions).

Layer	Healthy Mean $\pm$ SD (px)	SW (H)	DFU Mean $\pm$ SD (px)	SW (D)	t / p-value	Cohen's d	95% CI [DFU-H] (px)
Stratum Corneum (SC)	3.758 $\pm$ 0.328	p=0.0042 (non - Normal)	3.927 $\pm$ 0.514	p=0.0709 (Normal)	t = -1.0261 p = 0.3174	0.3914 (Small)	[-0.146, 0.459]
Epidermis (ED)	15.871 $\pm$ 5.837	p=0.0624 (Normal)	30.798 $\pm$ 10.773	p=0.0509 (Normal)	t = -4.4893, p = 0.0003*	1.7230 (Large)	[8.565, 21.272]

Dermis (DR)	37.343 ± 4.810	p=0.1769 (Normal)	26.750 ± 6.060	p=0.0055 (non-normal)	t = 5.1258 p = 0.0000*	-1.9363 (Large)	[-14.731, -6.754]
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* $p < 0.001$. SW = Shapiro-Wilk normality test. N = Normal, NN = Non-normal. CI = 95% bootstrap confidence interval. Highlighted rows indicate statistically significant between-group differences.

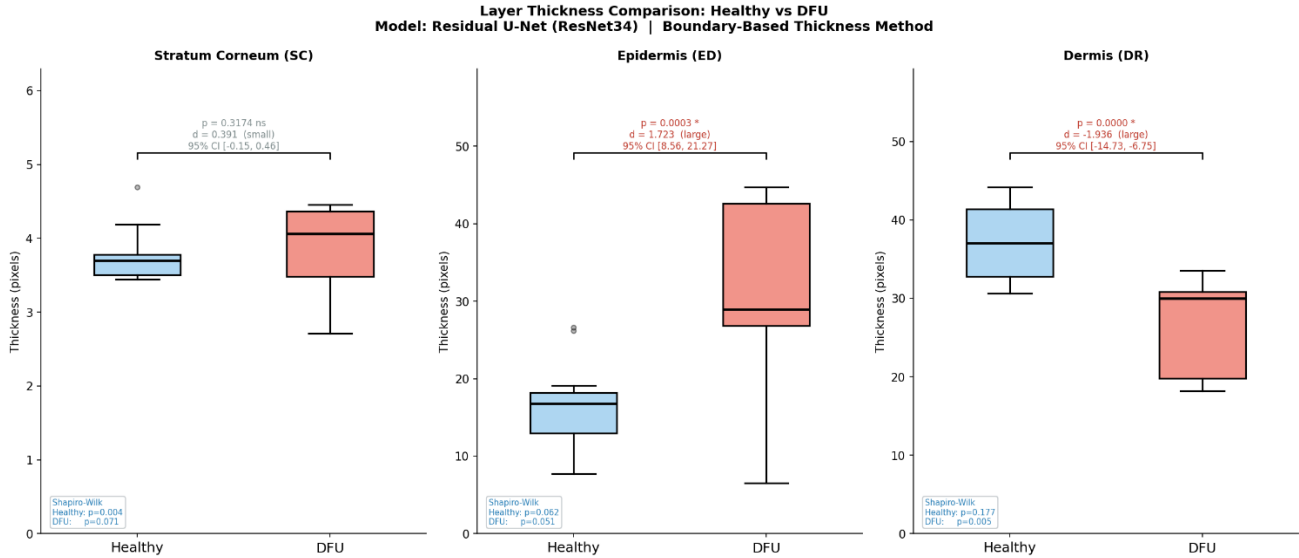


Figure 8. Layer thickness comparison between healthy and DFU cohorts for stratum corneum (SC), epidermis (ED), and dermis (DR). Boundary-based measurement from Residual U-Net (ResNet34)inspired Custom U-Net predictions.

The other important observation was epidermal thickening, with an average increase in the thickness of ED from 15.871 ± 5.837 px in healthy participants to 30.798 ± 10.773 px in DFU patients. The statistical analysis proved the significance of this finding as the obtained p-value ($p = 0.0003$) indicated a significant difference between the means of samples. The large Cohen's d of 1.723 with a 95% confidence interval entirely above zero ([8.565, 21.272]px) confirms that this is both a robust and clinically meaningful difference. Epidermal hypertrophy in diabetic plantar skin is attributed to hyperproliferative keratinocyte activity, impaired differentiation, and sustained inflammatory signaling under chronic glycemic stress [15].

As well as epidermis, dermis also became thinner, its mean thickness decreased by 30.0% to 26.75 ± 6.06 px comparing to control patients' value of 37.343 ± 4.81 px. Again, the very high reverse effect size (Cohen's $d = -1.9363$) along with a confidence interval that completely lies below zero ([-14.731, -6.754]px) supports the robustness and significance of this result. Dermal thinning of DFU skin is related to extracellular matrix degradation, collagen hypoproduction, and decreased fibroblast activity under long-term hyperglycemia [16]. It should be noted that Shapiro-

Wilk test for DFU dermis shows the existence of heterogeneous dermal injury between DFU patients (p-value = 0.0055).

Thickness of stratum corneum did not exhibit any statistical significance between the groups ($P=0.31914$, Cohen's $d=0.3914$). This suggests that the stratum corneum thickness is relatively intact even when deep-level alterations occur in DFUs. This is similar to previous studies which have shown that thickening of stratum corneum occurs in later stages of DFU development [5]. In combination, increased epidermis and decreased dermis thickness form the characteristic microstructural pattern of skin before the development of ulcers in diabetes patients.

3.4 Grad-CAM Explainability Analysis

Grad-CAM [31] algorithm was performed on the top-performing Residual U-Net (ResNet34) network to validate the model's attention patterns for all three segmentation classes. The grad-CAM saliency maps for SC, ED, and DR were created separately for images from both groups, i.e., healthy subjects and DFUs. Figure 9 displays the grad-CAM visualization results along with original OCT images for both subject groups.

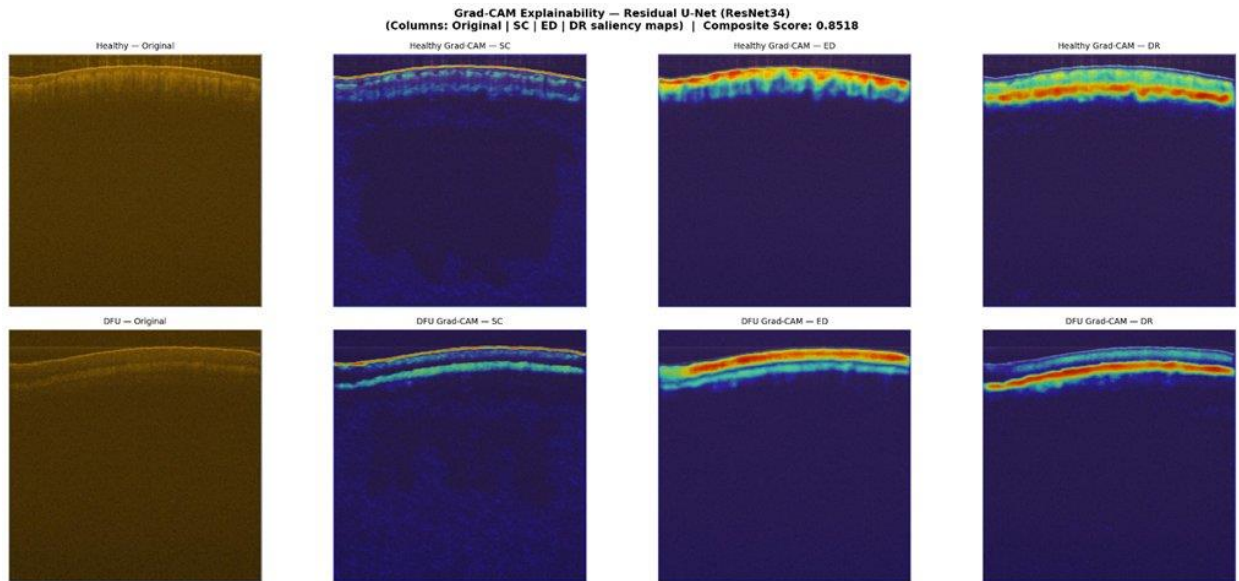


Figure 9. Grad-CAM explainability maps for Residual U-Net (ResNet34) on representative healthy (top row) and DFU (bottom row) OCT images. Columns show: original image, SC saliency map, ED saliency map, and DR saliency map. Composite Grad-CAM score: 0.8518. Warmer colors (red/yellow) indicate higher model attention.

The Grad-CAM results confirm that the model is able to pay attention to the appropriate skin layers within each class. For healthy patients, the activated areas are tightly confined to the boundaries of the skin layers, especially in the case of epidermis (ED), where the highest per-class

Dice score of 0.9204 was obtained in this population. On the other hand, the saliency map for the dermis (DR) class covers large parts of the dermal area but is accurately focused.

For the DFU patient group, the saliency map indicates that the model can still attend to the correct skin regions but exhibits less focused activation patterns, especially for the dermis class. Such an observation is expected due to the irregular and fragmented morphology of the dermis in DFUs and directly explains the lower Dice score of 0.7930, compared to the healthy group of 0.8958. The epidermis (ED) saliency maps for the DFU patients clearly highlight the thickened and altered epidermis layer due to epidermal hypertrophy in DFU pathophysiology.

Activation bands in saliency maps of the SC stratum in both cases have proper localization; they are narrow and confined only to the surface of the tissue, in correspondence with the anatomical location and the intact outlines of the thin layer of the SC stratum in both normal and DFU skin samples. Also, the explainability index of the integrated Grad-CAM score of 0.8518 as the spatial correlation coefficient between Grad-CAM activations and predicted segments outlines is proof of its clinical significance.

3.5 Cross-Validation and Generalization

In order to evaluate the generalization ability of the best performing Residual U-Net model, five-fold stratified cross validation analysis was done using the DFU train and val dataset. Table 4 presents fold-wise and summary statistics.

Table 4. Five-fold cross-validation results for Residual U-Net (ResNet34) on the DFU cohort.

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean	SD
Residual U-Net (ResNet34)	0.804	0.774	0.791	0.782	0.803	0.7909	0.0130

** Best-performing model. Each fold used 20 training epochs. Stratified splits preserved the ratio of DFU training and validation images.*

The Dice score mean for the model was found to be 0.7909 ± 0.0130 on all five folds of cross -validation. The consistency in terms of obtaining values between 0.77 and 0.80 shows that segmentation performance can be considered stable although at a comparatively lower level. The smaller standard deviation indicates the uniformity in the model's behavior across various partitions of data.

In spite of the low overall Dice score, the model does not appear to be overfitting to any particular fold. The Dice score obtained on each of the folds was similar and did not show much variation, suggesting the possibility of further improvements in the model's performance.

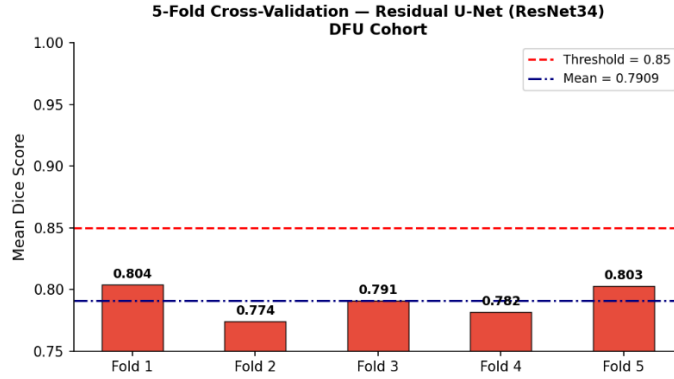


Figure 10. Five-fold cross-validation Dice scores for Residual U-Net (ResNet34) on the DFU cohort.

3.6 Clinical Classification Using Thickness Features

Logistic regression classifiers have been used to estimate the discriminatory power of thickness parameters based on boundaries through using a feature vector defined via combining the thickness values estimated for three boundaries predicted by Residual U-Net. The achieved classification accuracy with AUC-ROC of 0.9567 demonstrates high discriminatory ability of using just three thickness values (scalar feature vectors).

Epidermis and dermis boundaries provided the highest contribution to the classification process in comparison with stratum corneum, which can be explained by effect sizes obtained as a result of statistical testing. This result proves that there is the possibility of applying OCT technique along with automatic estimation of tissue thickness for developing a clinically applicable system which can discriminate patients with DFU and healthy subjects without performing manual segmentation of tissues by experts. From the practical perspective, such a system can be embedded into the point-of-care device which can identify the patients with unusual thickness distributions compared to the healthy population.

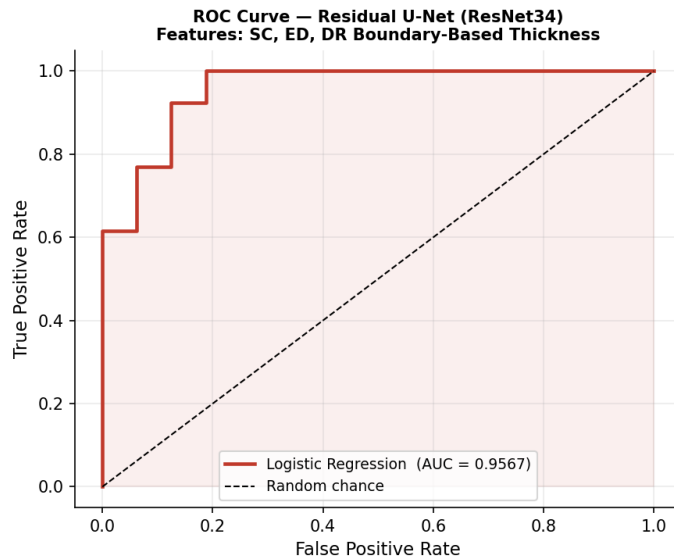


Figure 11. ROC curve for logistic regression clinical classification of healthy vs. DFU subjects using boundary-based thickness features (SC, ED, DR). AUC-ROC = 0.9567.

3.7 Computational Complexity and Deployment Characteristics

Computational performance attributes of all eleven models can be found in Table 5. Each model has been tested on a machine with an NVIDIA Tesla T4 GPU that had a memory capacity of 15.6 GB. All models had the effective batch size of 4 consistently through the experiment; for U-Net++, Attention U-Net, and Dense U-Net, an actual batch size of 1 was set with 4 accumulations because of their relatively large memory consumption. Among the predefined benchmark models, Swin-UNet required 250.1 ms due to the computational overhead of shifted-window attention mechanisms, while MedSAM-UNet required 3492 ms owing to the SAM ViT-B encoder operating at 1024×1024 resolution, necessitating server-level GPU deployment rather than a standard clinical workstation. This substantial inference cost highlights a key practical limitation of foundation model approaches for real-time clinical deployment despite competitive segmentation accuracy.

Table 5. Computational complexity and deployment characteristics of all eleven architectures.

Model	Params (M)	Train Time (hr)	Inference (ms)	Batch Config	Deployment
U-Net (ResNet34)	33.71	1.29	129.8	BS=4	Clinical workstation
U-Net++ (ResNet34)	40.03	1.24	143.3	BS=1, Accum=4	Clinical workstation
U-Net++ (EfficientNet-B4)	33.84	1.58	128.1	BS=4	Clinical workstation
DeepLabV3+ (ResNet34)	29.26	1.66	162.2	BS=4	Clinical workstation
Attention U-Net (ResNet50)	34.24	1.15	134.5	BS=1, Accum=4	Clinical workstation
Residual U-Net (ResNet34) *	33.71	1.96	128.2	BS=4	Clinical workstation
Dense U-Net (DenseNet121)	37.21	1.23	147.8	BS=1, Accum=4	Clinical workstation
U-Net (EfficientNet-B4)	8.67	0.77	71.1	BS=4	Clinical workstation
U-Net (VGG-16)	58.86	2.41	151.0	BS=4	Clinical workstation
Swin-UNet (Swin-Tiny)	27.1	4.2	250.1	BS=4	Clinical workstation
MedSAM-UNet (SAM ViT-B)	19.8	3.8	3492	BS=4	Server / GPU cluster

* Best-performing model. BS = batch size; Accum = gradient accumulation steps. Training times are approximate. Inference times measured with synchronized GPU timing.

The inference rate for Residual U-Net(ResNet34) was 128.2 ms , putting it in the feasible inference time range yet lower in speed than the light models. On the other hand, the base model (U-Net ResNet34) had a comparable inference time of 129.8 ms, demonstrating that the use of scaled residual connections in no way impacts the inference time adversely. In terms of the inference

efficiency, U-Net with EfficientNet-B4 encoder performed the best in the group. In particular, U-Net (EfficientNet-B4) demonstrated relatively low inference time of 71.1 ms along with the lowest number of parameters (8.67M).

At the same time, U-Net++ and Dense U-Net as well as U-Net (VGG-16) needed notably more inference time, amounting from 143 ms up to 151 ms correspondingly. Also among the custom architecture variants, DeepLabV3+ (ResNet34) had the highest inference time per image 162.2 ms, which might be attributed to its dilated convolutional pyramid and context aggregation design and attention U-Net (ResNet50) had an inference rate of 134.5 ms .

4. Clinical Significance

The modifications noted in the structure of skin, such as thickened epidermis and thinned dermis in DFUs, are of great clinical significance because they suggest the likelihood of ulcers in patients suffering from diabetes foot syndrome and ways to prevent this variant of the condition. Epidermal thickening occurs through hyperproliferation of keratinocytes, reduced protection of the epidermis layer, and continuous inflammation caused by high blood glucose. Thickened epidermis on the soles of the foot arises due to increased mechanical stress that is an important factor in mechanical risk to the development of DFUs in presence of diabetic neuropathy [17].

Thin dermis is attributed to degradation of the extracellular matrix, diminished production of collagen fibers, dysfunctional activity of fibroblasts, and increased accumulation of advanced glycation end products in the dermis layer. This affects the capacity of the skin to withstand mechanical pressure. Mechanical stress influences the formation of ulcers in diabetics directly in a number of biomechanical ways [18]. Abnormality in the dermis layer of the DFUs, as seen from Shapiro-Wilk distribution ($p = 0.0055$), exacerbates the structural modification in the dermis layer of DFUs.

The Grad-CAM analysis provides additional clinical value by confirming that model predictions are driven by attention to anatomically relevant layer boundaries rather than image artefacts or background features. This explainability is essential for building clinician trust in AI-assisted diagnostic tools and for identifying failure modes in edge cases.

The result of 0.9567 achieved by the logistic regression classifier model with respect to thickness features is indicative of the fact that OCT structure assessment is valuable enough to influence the process of medical decision-making, despite the absence of a specialist for image analysis. The process that involves utilizing OCT imaging, analysis of layer thicknesses, and referral of patients with abnormal distributions to podiatrists can successfully identify patients whose layer thicknesses differ significantly from normal distribution and can thus proceed to preventative care prior to the development of ulcers clinically.

With respect to telemedicine, the inference delay for each OCT image when using Residual U-Net is 18.2 ms. It means that microstructural OCT evaluation can be done in real-time conditions. Utilization of REST API, such as TorchServe, for delivery of segmented OCT images and layer thicknesses may enable medical professionals to evaluate OCT structures in the same way as a podiatrist.

5. Limitations and Future Work

Several limitations associated with the present study must be taken into account. Firstly, the size of the image pool, including both images of healthy people and those with DFU (397 and 326 images, respectively), is rather small for training and testing eleven DL algorithms. While consistent results and p-values have been achieved, further testing of the model using multi-center OCT image pools is necessary prior to practical application. Secondly, as mentioned above, OCT imaging techniques used here include only one modality, hence hardware-related confounding variables. Thirdly, all the nine BARS-UNet architectures used in this paper are initialized randomly, which means that no pretrained weight was used in the process. This approach guarantees fair comparisons among different DL algorithms. However, it may negatively affect the overall predictive ability of the network compared to when it is initialized using pretrained weights trained on ImageNet, especially if few training data are available. Fourthly, as test and validation sets represent the same pool, this restricts variance estimation by means of cross-validation. Finally, logistic regression was used solely based on thickness features.

Several avenues are open for further investigation. The enlargement of the training set by employing the multi-center approach, adding data to the existing data set, and implementing semi-supervised learning on the unlabelled OCT data will be considered. The fine-tuning of the customized BARS-UNet algorithm through the domain adaptation technique, wherein pre-training is conducted on large datasets, will also be investigated. The segmentation of OCT images to a 3D volume will allow three-dimensional structural analysis. The clinical study on the association between automated thickness measurements and DFU staging and wound healing will be needed.

6. Conclusion

The research provided an extensive deep learning-driven methodology for automatic multi-class segmentation and boundary thickness estimation analysis for skin OCT images from normal subjects and DFU patients. Total of 11 models including two predefined and nine custom segmentation models were analyzed in the context of a consistent custom BARS-UNet approach that utilizes residual connectivity, squeeze-and-excitation attention mechanism, and a separate boundary-driven feature pathway uniformly applied to all model variations.

The Residual U-Net (ResNet34) variation, with its improved weighted residuals and optimized skip connections, delivered the highest DFU result with a test Dice score of 0.8518 and a healthy population Dice score of 0.8998. Individual class analysis revealed that the epidermis layer was most precisely segmented among the healthy population (Dice = 0.9204) while the dermis layer showed the worst performance deterioration when analyzing DFU (Dice = 0.7930).

Thickness analysis through boundary-based approach showed that DFU patients had thicker epidermis ($p=0.0003$, $d=1.723$; large effect) and thinner dermis ($p=0.00001$, $d=-1.9363$; large effect) compared to healthy subjects, whereas the thickness of the stratum corneum did not change. Grad-CAM analysis confirmed that model attention is correctly localized to clinically relevant skin layer boundaries, supporting the explainability and trustworthiness of the framework for clinical deployment. Classification results of the proposed logistic regression model based on the input data of thickness metrics were excellent, with a ROC-AUC value of 0.9567. This demonstrated the utility of the automated boundary-based thickness profiling for discriminating healthy from at-risk patients. Results of five-fold cross validation provided additional proof of model robustness with a mean DFU Dice of 0.7909 ± 0.0130 .

Overall, the suggested methodology provides a practical, reliable and interpretable tool for assessing alterations in microstructure of skin associated with DFU pathology. With the inference speed being as high as 128.2 ms and capability of working in clinical GPUs, this framework can be used as an auxiliary diagnostic component of DFU screening algorithms. In the future, this algorithm could be further validated across multiple centers, and extended to incorporate three-dimensional OCT and clinical outcome data.

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